

1. Background

1.1 Enhancement and Depletion MOSFET Types

MOSFETs are voltage-controlled transistors where the gate is isolated from the channel by a thin oxide layer - producing virtually zero DC gate current.

Enhancement-mode NMOS (used in this lab - BS170): no channel exists at $V_{GS} = 0$. Applying $V_{GS} > V_t$ induces an electron inversion layer (channel). The device is **normally OFF**.

$$I_D = \frac{k_n}{2}(V_{GS} - V_t)^2 \quad (\text{saturation})$$

Depletion-mode NMOS: a built-in channel exists at $V_{GS} = 0$ - **normally ON**. A negative V_{GS} depletes the channel. Same square-law applies with $V_t < 0$.

Property	Enhancement	Depletion
Channel at $V_{GS} = 0$	No	Yes
Normally	OFF	ON
V_t (N-ch)	Positive	Negative
Use	Digital, switching, amplifiers	Current sources, analog

Both types exist in P-channel versions with all polarities reversed.

1.2 Why the Distinction Between Drain and Source?

Structurally the drain and source are **identical N⁺ diffusions** - the device is geometrically symmetric. The distinction is purely by **circuit biasing**:

1. **The source is the lower-potential terminal** (NMOS). V_{GS} and V_{DS} are always referenced to it; swapping them changes the operating point entirely.
2. **The body/substrate is tied to the source** to keep the body-source junction reverse biased. Reversing it causes body-diode conduction and loss of gate control.
3. In saturation the channel **pinches off at the drain end** - this rectifying asymmetry enables amplification and switching.

1.3 MOSFET as Amplifier and Switch

Region	Condition	Use
Cut-off	$V_{GS} < V_t$	Switch OFF - $I_D \approx 0$
Triode (linear)	$V_{GS} > V_t, V_{DS} < V_{GS} - V_t$	Switch ON - small $R_{DS,on}$
Saturation	$V_{GS} > V_t, V_{DS} \geq V_{GS} - V_t$	Amplifier - $I_D = \frac{k_n}{2}(V_{GS} - V_t)^2$

As an amplifier (saturation): $i_d = g_m v_{gs}$, $g_m = k_n(V_{GS,Q} - V_t)$, CS gain $A_V = -g_m R_D$.

As a switch (cut-off $\leftrightarrow$ triode/saturation): gate voltage alone toggles the channel; zero steady-state gate current is the key advantage over the BJT.

1.4 MOSFET vs. BJT - Unique Properties

Property	MOSFET	BJT
Gate/base current	≈ 0 (oxide insulated)	$I_B = I_C/\beta \neq 0$
Input impedance	Virtually infinite	Finite ($r_\pi \sim \text{k}\Omega$)
Control	Voltage (V_{GS})	Current (I_B)
Static power	Negligible	Base current dissipates power
VLSI scaling	Excellent (nm gate today)	Difficult below ~ 100 nm
CMOS logic	Zero static power	No equivalent
$1/f$ noise	Higher	Lower

Why MOSFET dominates VLSI: Zero gate current allows billions of gates; CMOS pairs draw current only during switching; gate oxide scales with lithography; simpler fabrication.

2. Gate-Drain Characteristics

(a) Simulation Result

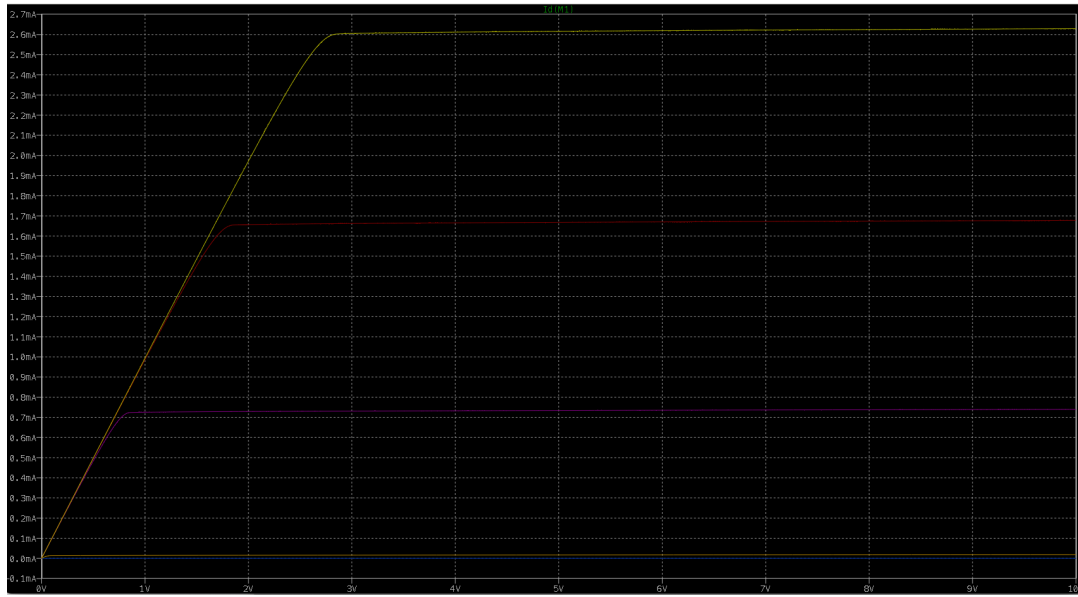


Figure 1: I_D vs. V_{DS} for $V_G = 0.5$ V

We can clearly see the saturation behavior of the MOSFET: beyond some point I_{DS} depends only on V_{GS} . The bigger V_{GS} is, the higher the saturation current. The triode region is the rising linear part, followed by a horizontal region (saturation) with a tiny Early-effect slope.

(b) Threshold Voltage Extraction

From the simulation the first conducting curve appears at an applied gate voltage of $V_g = 3$ V. Because the source sits on $R_2 = 1$ k Ω , the true gate-source voltage is $V_{GS} = V_g - I_D R_2$ (e.g. $V_{GS_3} = 3 - 0.73 = 2.27$ V, $V_{GS_4} = 4 - 1.67 = 2.33$ V). Taking the two lowest conducting curves and eliminating K through their ratio:

$$\begin{aligned}
 I_{DS_4} &= K(V_{GS_4} - V_t)^2 = 1.65[mA] \\
 I_{DS_3} &= K(V_{GS_3} - V_t)^2 = 0.73[mA] \\
 2.26 &= \frac{(V_{GS_4} - V_t)^2}{(V_{GS_3} - V_t)^2} \\
 \Rightarrow 2.26(V_{GS_3}^2 - 2V_{GS_3}V_t + V_t^2) &= (V_{GS_4}^2 - 2V_{GS_4}V_t + V_t^2) \\
 \Rightarrow 2.26V_{GS_3}^2 - V_{GS_4}^2 - 2(2.26V_{GS_3} - V_{GS_4})V_t + 1.26V_t^2 &= 0 \\
 \text{Since: } V_{GS_4} &= 2.33[V], V_{GS_3} = 2.27[V] \\
 \boxed{V_{t_1} = 2.29[V], V_{t_2} = 2.15[V]} \\
 \text{under the saturation assumption } \Rightarrow \boxed{V_t \approx 2.15[V]}
 \end{aligned}$$

Which aligns with the typical threshold voltage of $V_t \approx 2.1$ V from the provided datasheet.

(c) Operating Regions

Region	Graph location	Condition
Cut-off	$I_D \approx 0$, $V_{GS} = 0, 1, 2$ [V]	$V_{GS} < V_t$
Triode	Steep rising slope, low V_{DS}	$V_{DS} < V_{GS} - V_t$
Saturation	Flat plateau, high V_{DS}	$V_{DS} \geq V_{GS} - V_t$

3. The MOSFET as a Switch

(a) Transient - Non-Symmetric Sine (0-5 V)

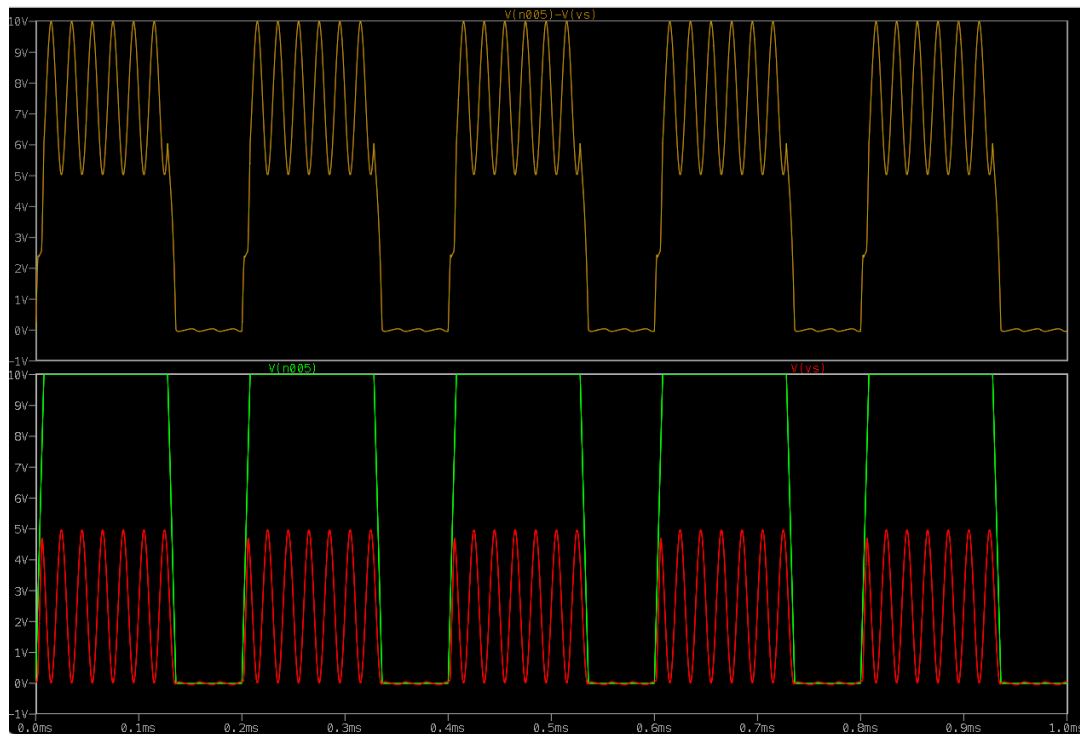


Figure 2: V_G (Green) and V_S (Red) ; V_{GS} (Orange).

We can see that on the high peaks of the clock the transistor is ‘open’ thus there is a voltage drop on the source resistor and V_S is a positive sine (low voltage drop over the transistor means it is saturated). On the other hand, on the low peak of the clock the transistor is closed, no current flows through the source resistor, thus V_S is only leakage. The AC signals are big enough to change the transistor’s operating point.

(b) Operating Regime

Gate	V_{GS}	Region	Switch
HIGH (10 V)	$\gg V_t$	Saturation	Open $I_D \gg 0$
LOW (0 V)	$< V_t$	Cut-off	Closed, $I_D \approx 0$

The saturation region is the correct one since the gate voltage is always bigger than the drain voltage.

(c) On-Resistance $R_{DS,on}$

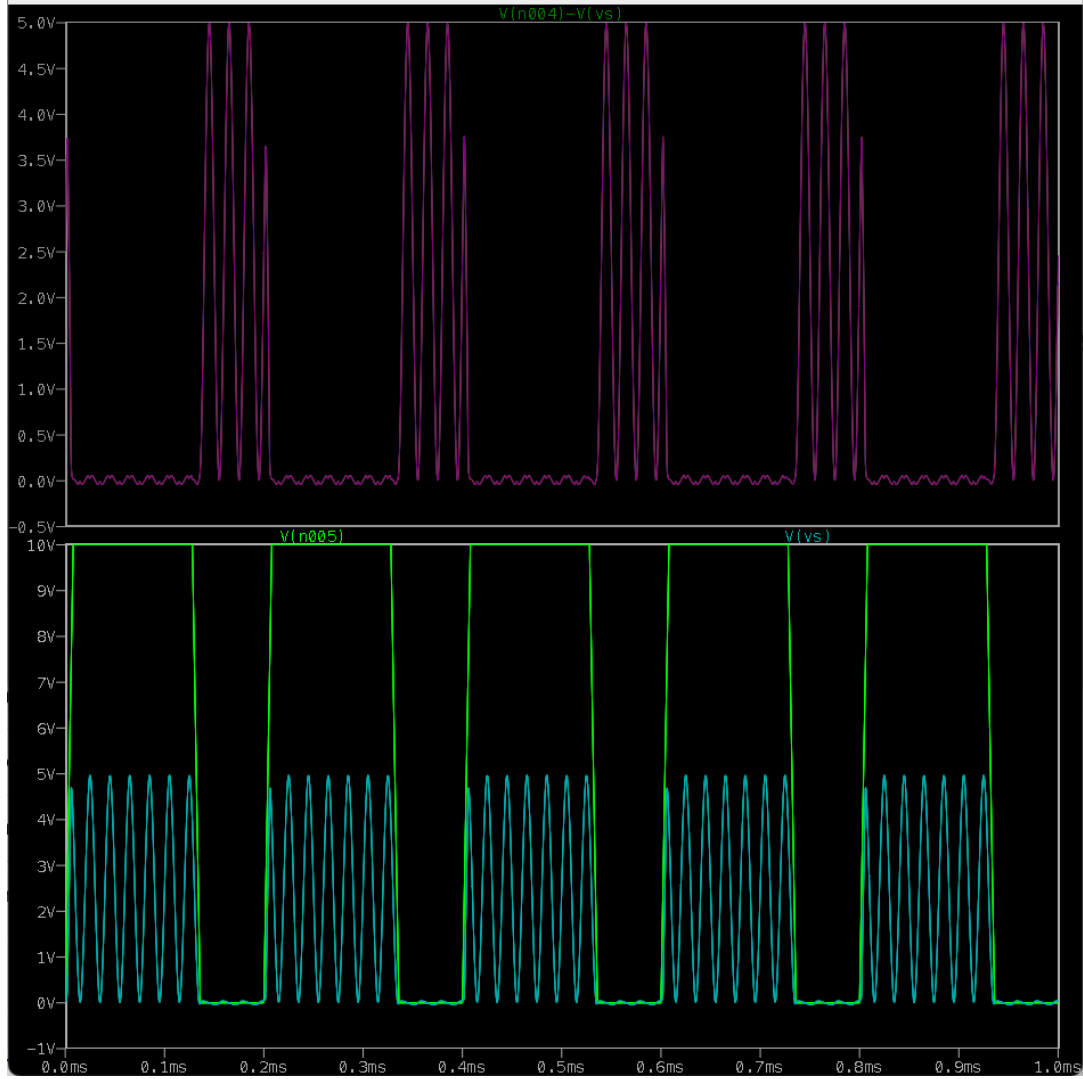


Figure 3: R_{DS} extraction; on-state: $V_{DS} \approx 0\text{ mV}$, $I_D \approx 5\text{ mA}$.

$$R_{ds} = \frac{V_{DS}}{I_D} = \frac{V_D - V_S}{\frac{V_S}{R_S}} = R_S \left(\frac{V_D}{V_S} - 1 \right)$$

Since the O.P. is changed, we choose one peak point of V_S on the graph. Assuming the delay is negligible, V_D at the peaks is 5 V. Thus:

$$R_{ds}|_{on} = R_S \left(\frac{V_D}{V_S} - 1 \right) = R_S \left(\frac{5}{4.972} - 1 \right) = 5.63\Omega$$

(d) Symmetric Sine $\pm 2.5 \text{ V} / 50 \text{ kHz}$

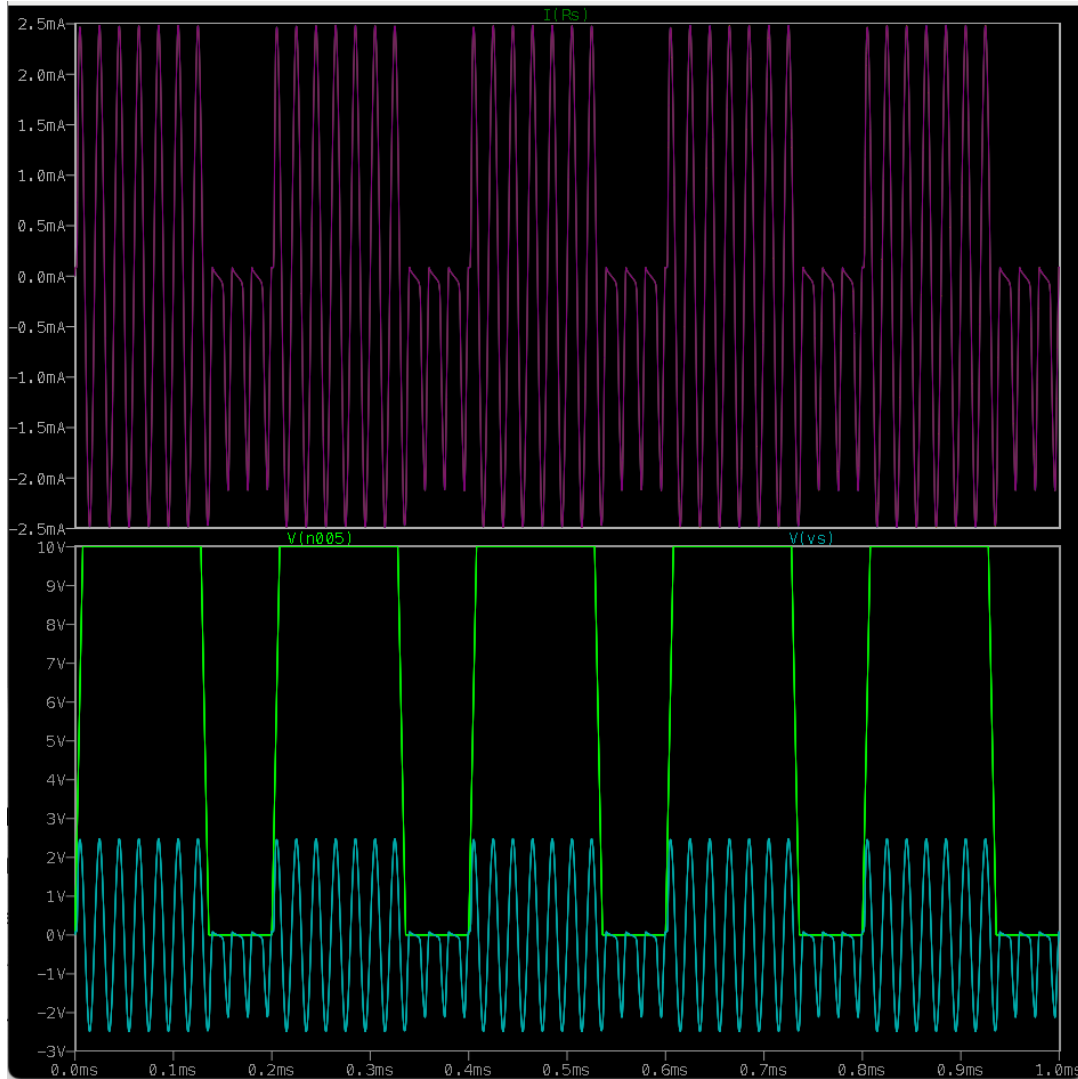


Figure 4: V_S with symmetric $\pm 2.5 \text{ V}$ sine; negative half-cycles pass via body diode regardless of gate state.

Explanation: For $V_2 > 0$ and gate HIGH the MOSFET conducts (saturation) since $V_{GS} \gg V_{DS} - V_t$. For $V_2 < 0$ (and gate HIGH) it stays the same, however the current through the resistor flips direction. For V_G LOW the transistor is closed. However, since V_{DS} keeps shifting, we can see a negative current flow (S to D), so it has some resistance. Since $V_S = V_2 \frac{R_S}{R_S + R_{OFF}}$, $R_{ds|OFF} = R_S(\frac{V_2}{V_S} - 1) = 1(\frac{2.5}{2.18} - 1) = 147 \Omega$. Generally speaking, from the geometric symmetry of an NMOS transistor, we can conduct current in both directions, but it will be less efficient due to the high body effect.

4. Common-Source MOSFET Amplifier

(a) DC Bias - Personalized Values ($R_f = 9.6 \text{ k}\Omega$, $R_2 = R_3 = 6.5 \text{ k}\Omega$)

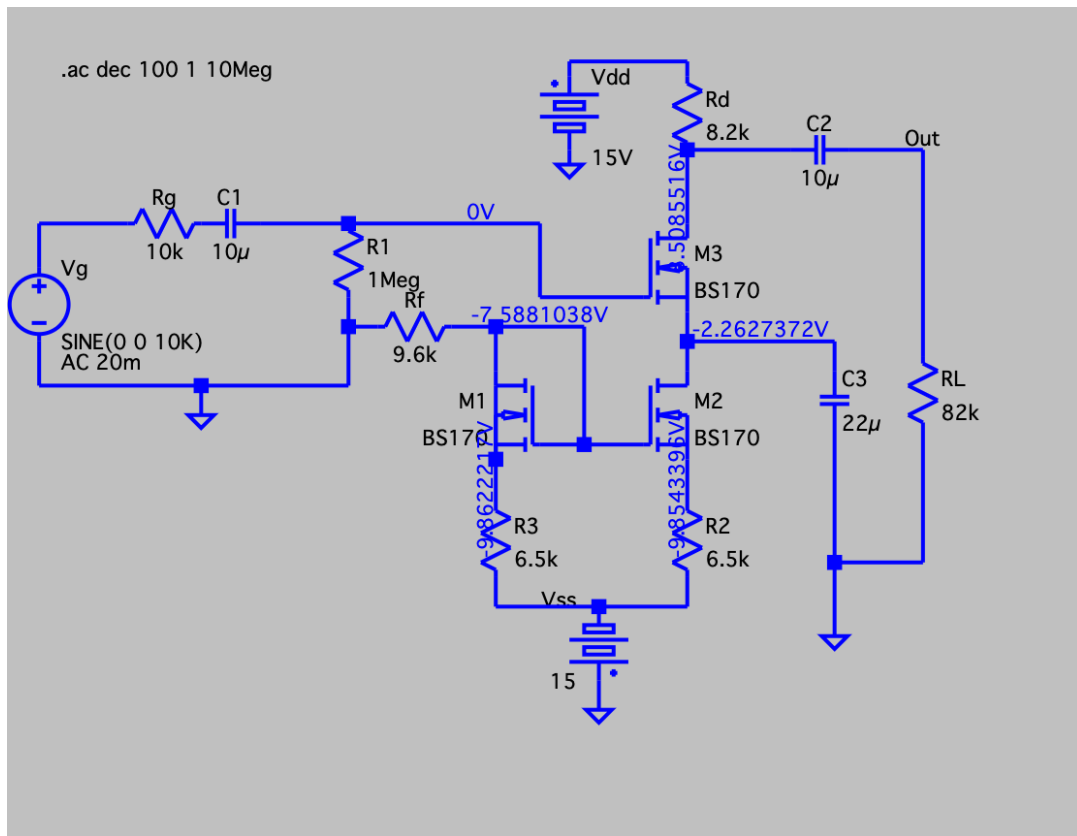


Figure 5: DC operating point - personalized values; $I_D \approx 1 \text{ mA}$.

(b) DC Bias - Standard Values, Operating Points and g_{m_3}

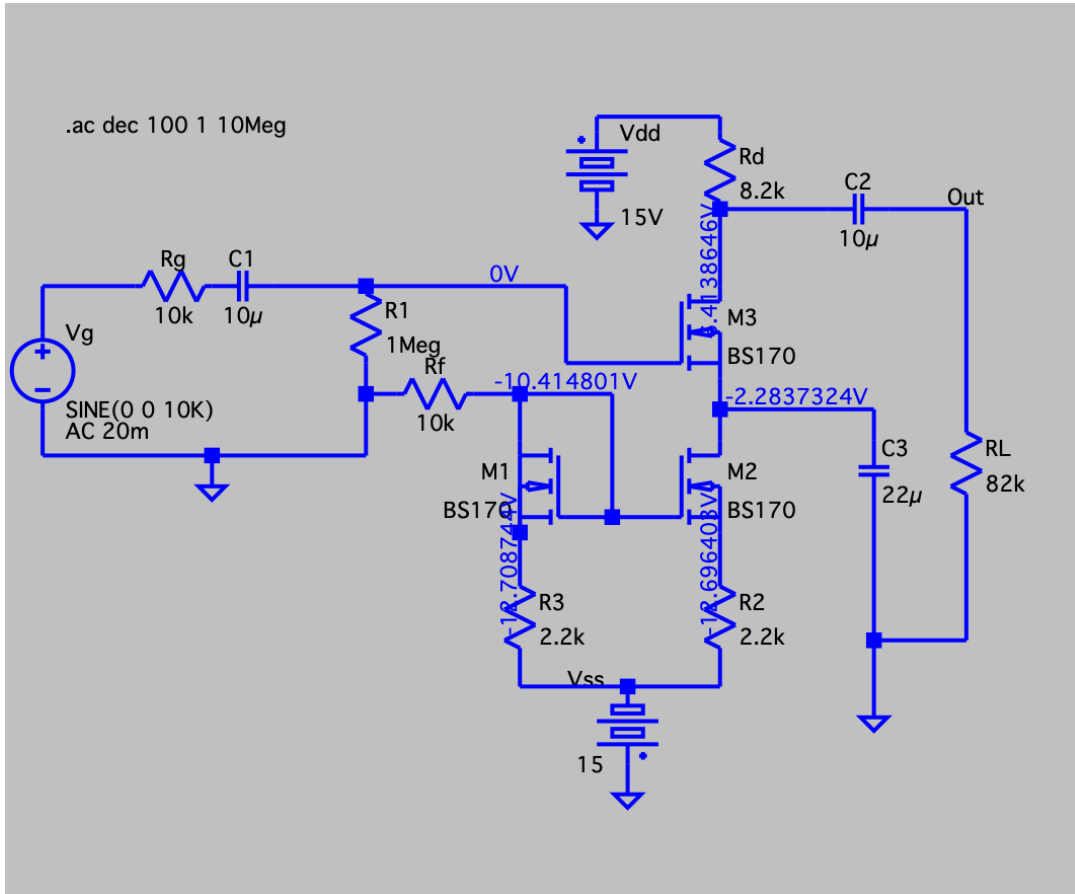


Figure 6: DC measurement - $I_D(M_1) = 1.041 \text{ mA}$, $I_D(M_3) = 1.047 \text{ mA}$

Bias verification:

$$V_{D3} = V_{DD} - I_{D3} \cdot R_d = 15 - 1.047 \times 8.2 = \mathbf{6.41 \text{ V}}$$

$$V_{S1} = V_{SS} + I_{D1} \cdot R_3 = -15 + 1.041 \times 2.2 = \mathbf{-12.71 \text{ V}}$$

Operating modes:

Transistor	$V_{GS} \text{ [V]}$	$V_{DS} \text{ [V]}$	Mode
M1	2.29	2.29	Triode ($V_{DS} \ll V_{GS} - V_t$), Active resistor
M2	2.29	10.43	Saturation ($V_{DS} \gg V_{GS} - V_t$)
M3	2.28	4.13	Saturation ($V_{DS} \gg V_{GS} - V_t$)

Transconductance $g_{m,M3}$:

$$g_{m,M3} = \frac{2 I_{D3}}{V_{GS,M3} - V_t} = \frac{2 \times 1.047}{2.28 - 2.15} \approx \mathbf{16.1 \text{ mS}}$$

(c) Small-Signal Equivalent Circuit

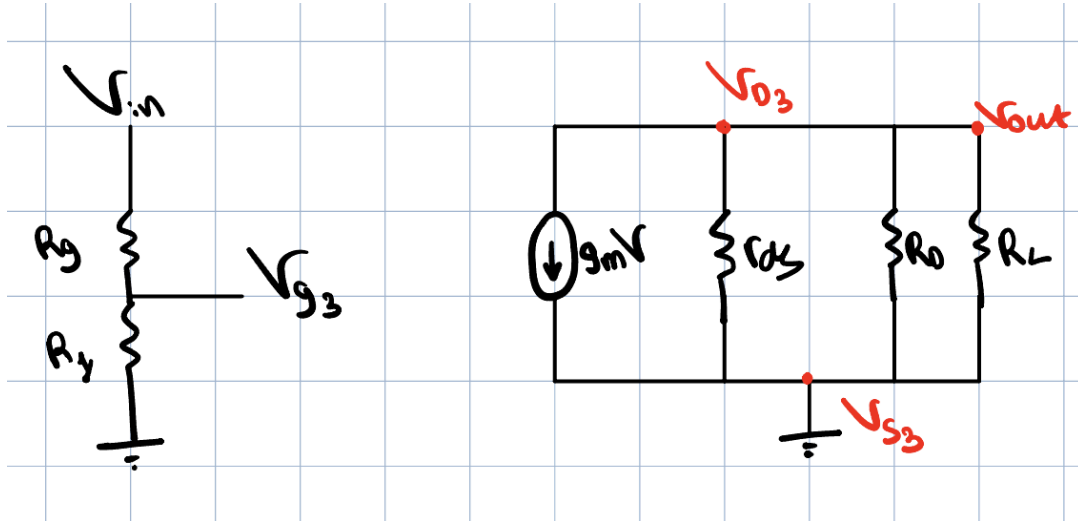


Figure 7: Small-signal model: $g_m v_{gs}$ source with $r_{ds} \parallel R_d \parallel R_L$ at the output

$$V_{out} = -g_{m,M3} \cdot v_{gs,M3} \cdot (r_{ds} \parallel R_d \parallel R_L)$$

The current mirror formed by M1 and M2 is grounded by capacitor C_3 .

(d) Purpose of Capacitor C_3

C_3 (22 μ F) bypasses the source-degeneration resistor R_2 of M2.

- **DC:** C_3 open - R_2 stabilises the bias point via negative feedback.
- **AC:** C_3 shorts R_2 , restoring full transconductance and preventing gain loss:

$$A_V \Big|_{\text{with } C_3} = -g_m R_d \quad \text{vs.} \quad \frac{-g_m R_d}{1 + g_m R_2} \Big|_{\text{without}}$$

For $g_m = 16.1$ mA/V, $R_2 = 2.2$ k Ω : without C_3 the gain drops by a factor of 36 (−31 dB). Low-frequency cutoff: $f_L = 1/(2\pi C_3 R_2) \approx 3.3$ Hz - well below the signal band.

(e) Frequency Response - $R_f = 10\text{ k}\Omega$ vs. $R_f = 16\text{ k}\Omega$

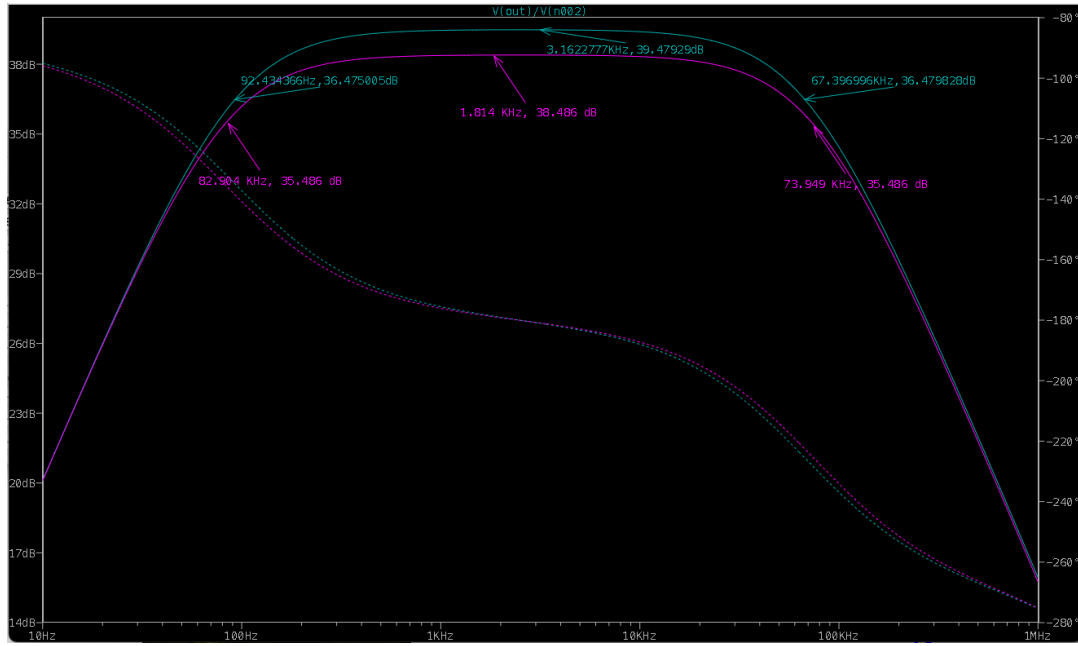


Figure 8: Bode plot - $R_f = 10\text{ k}\Omega$: 39.5 dB, BW $\approx 67\text{ kHz}$; $R_f = 16\text{ k}\Omega$: 38.5 dB, BW $\approx 74\text{ kHz}$.

R_f	Mid-band [dB]	f_L [Hz]	f_H [kHz]	BW [kHz]
10 k Ω	39.48	92.4	67.4	67.3
16 k Ω	38.49	82.9	73.9	73.8

From the results we can see two main things. First, the gain-BW tradeoff applies to the circuit, as expected. Second, as we keep increasing R_f we change the OP such that the current through M3 gets smaller. If we keep increasing it, M3 will shift out of saturation, reducing the gain proportionally to $g_{m,M3}$.

(f) Output Resistance

$$R_{out} = R_L \Big|_{V_{load}=V_{OC}/2}$$

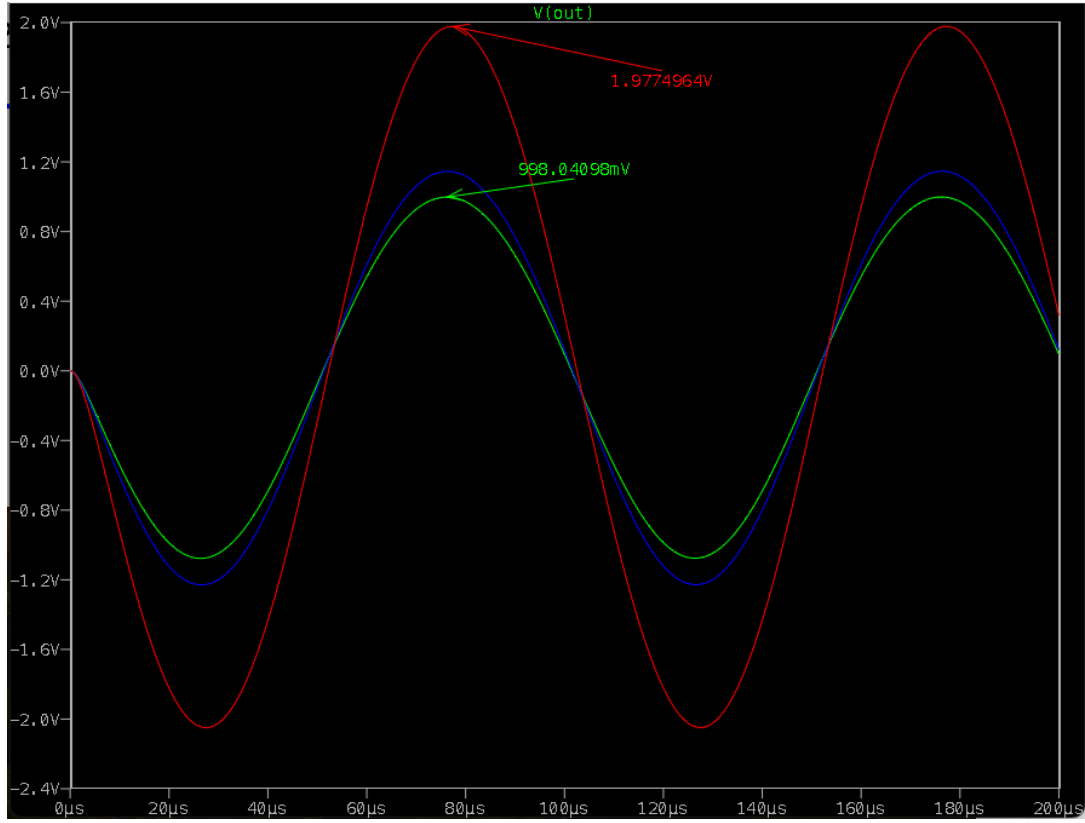


Figure 9: R_{out} measurement - $V_{OC} = 1.977\text{ V}$; half-voltage (998 mV) occurs at $R_L \approx R_{out} \approx 7.4\text{ k}\Omega$.

$$V_{OC} = 1.986\text{ V}, \quad V_{half} = 994\text{ mV} \Rightarrow \boxed{R_{out} \approx R_L|_{V=\frac{V_{OC}}{2}} \approx 7.4\text{ k}\Omega}$$

Theoretical estimate from the small-signal model:

$$R_{out} \approx R_d \parallel r_{ds} = 8.2\text{ K} \parallel 50\text{ G} \approx 8.19\text{ K}\Omega$$

This variation happens because, in the simulation, the source is not 100% grounded, so some resistance ‘propagates’ with the gm_{M3} factor from the source to the drain.

(g) Purpose and Applications

This self-biased three-MOSFET CS amplifier with R_f feedback delivers:

- High gain ($\sim 40\text{ dB}$), wide bandwidth ($\sim 70\text{ kHz}$), programmable by R_f alone.
- Self-stabilising DC bias - no external reference needed.
- Very high input impedance.

Applications: photodiode transimpedance amplifier (TIA), sensor signal conditioning, high-impedance instrumentation front-end, audio pre-amplifier stage.